

SCHOOL ERP SYSTEM

Functional Workflow & Module Dependency Document

Project Information

Project Name: School ERP / School Management System

School: Prajnanabodhini English Medium School & Jr. College, Shirgaon / Chikhali

Technology Stack

- Laravel 13.x
- PHP 8.3
- MySQL / MariaDB
- Blade Templates
- Tailwind CSS

Database

- school_management
-

1. SYSTEM OVERVIEW

The School ERP is designed to manage the complete academic workflow from student admission through marks entry, result generation, and report card printing.

The current implementation focuses on:

1. Authentication
 2. Master Data Management
 3. Student Management
 4. Teacher Allocation
 5. Marks Management
 6. Result Processing
 7. Report Card Generation
-

2. COMPLETE WORKFLOW

```
Login
↓
Academic Year Selection
↓
Master Setup
↓
Student Admission
↓
Teacher Allocation
↓
Skill Subject Allocation
↓
Marks Entry
↓
Marks Validation
↓
Marks Locking
↓
Result Generation
↓
Report Card Printing
```

3. MODULE DEPENDENCY FLOW

```
Academic Year Master
↓
Standard Master
↓
Division Master
↓
Subject Type Master
↓
Subject Master
↓
Exam Master
↓
Student Records
↓
Teacher Class Allocation
↓
```

Teacher Subject Allocation
↓
Student Skill Subject Allocation
↓
Marks Entry
↓
Teacher Marks Status
↓
Result Generation
↓
Report Card

4. AUTHENTICATION MODULE

Purpose

Controls access to the ERP system.

Features

- Login
- Registration
- Dashboard Access

Database Table

users

User Types

- Administrator
- Teacher

5. ACADEMIC YEAR MASTER

Purpose

Defines the active academic session.

Examples

2025-2026
2026-2027

Database Table

academic_years

Dependency

Required by all modules.

6. STANDARD MASTER

Purpose

Defines school classes.

Examples

1st Standard
2nd Standard
3rd Standard
...
10th Standard

Database Table

standards

Dependency

Requires Academic Year.

7. DIVISION MASTER

Purpose

Defines class divisions.

Examples

A
B
C

Database Table

divisions

Dependency

Requires Standard Master.

Example:

5th Standard
├─ A
├─ B
└─ C

8. SUBJECT TYPE MASTER

Purpose

Categorizes subjects.

Examples

Academic
Skill
Optional

Database Table

subject_types

Dependency

Used by Subject Master.

9. SUBJECT MASTER

Purpose

Stores all school subjects.

Examples

English
Marathi
Hindi
Sanskrit
Mathematics
EVS
Computer
Robotics
Physical Education
Acting
Music
Drawing

Database Table

subjects

Dependency

Requires Subject Type.

Example:

English → Academic

Acting → Skill

Music → Skill

10. EXAM MASTER

Purpose

Defines examinations and marking schemes.

Examples

Unit Test 1

Unit Test 2

Semester Exam

Annual Exam

Database Table

exam_masters

Configuration

Each exam may contain:

Theory Marks

Theory Passing Marks

Oral Marks

Oral Passing Marks

Practical Marks
Practical Passing Marks

Dependency

Requires Subject Master.

11. STUDENT RECORDS MODULE

Purpose

Stores complete student information.

Database Table

students

Information Stored

Admission Number
Student Name
Date of Birth
Gender
Mobile Number
Address
Photo
Standard
Division
Academic Year

Dependency

Requires:

- Academic Year
 - Standard
 - Division
-

12. TEACHER CLASS ALLOCATION

Purpose

Assigns a class teacher.

Database Table

teacher_class_allocations

Example

Teacher A → 5th A

Teacher B → 6th B

Dependency

Requires:

- Teacher
- Standard
- Division
- Academic Year

13. TEACHER SUBJECT ALLOCATION

Purpose

Assigns subjects to teachers.

Database Table

teacher_subject_allocations

Example

Teacher A → Mathematics → 5th A

Teacher B → English → 5th A

Dependency

Requires:

- Teacher
- Subject
- Standard
- Division
- Academic Year

Importance

Marks Entry depends completely on this module.

Without subject allocation, marks cannot be entered.

14. STUDENT SKILL SUBJECT ALLOCATION

Purpose

Assigns one optional skill subject to each student.

Database Table

student_skill_subjects

Examples

Acting
Music
Drawing
Dance

Dependency

Requires:

- Student
- Skill Subject
- Academic Year

Usage

Used during Report Card generation.

Only one selected skill subject appears on the final report card.

15. MARKS ENTRY MODULE

Purpose

Allows teachers to enter examination marks.

Controller

```
app/Http/Controllers/MarksEntryController.php
```

Workflow

Teacher Login

↓

Select Exam

↓

Select Subject Allocation

↓

Load Student List

↓

Enter Marks



Save Marks

Database Table

```
student_marks
```

Dependency

Requires:

- Student Records
- Exam Master
- Teacher Subject Allocation
- Subject Master

16. MARKS ENTRY VALIDATION

Purpose

Prevents duplicate marks submission.

Validation Logic

Before saving:

```
StudentMark::exists()
```

If marks already submitted:

```
UNIT TEST 1 Teacher Assignment is already completed.
```

Marks entry is blocked.

Status

Working Successfully.

17. TEACHER MARKS STATUS MODULE

Purpose

Tracks completion and locking of marks entry.

Database Table

```
teacher_marks_status
```

Function

```
Pending  
Completed  
Locked
```

Dependency

Requires Marks Entry.

18. RESULT PROCESSING MODULE

Purpose

Generates examination results.

Processing Includes

```
Subject Totals  
  
Pass / Fail  
  
Percentage
```

Overall Result

Database Tables

student_results

student_result_details

Dependency

Requires:

- Marks Entry
- Teacher Marks Status

19. REPORT CARD MODULE

Purpose

Generates final printable report card.

Controller

app/Http/Controllers/Administrator/ReportCardController.php

View

resources/views/report-card/index.blade.php

Dependency

Requires:

- Student Results
- Student Result Details
- Student Skill Subjects

20. REPORT CARD SUBJECT DISPLAY LOGIC

The report card displays:

Academic Subjects

```
subject_type_id <> 2
```

Always Include

```
COMPUTER  
ROBOTICS  
PHYSICAL EDUCATION
```

Skill Subject

Display only ONE selected skill subject from:

```
student_skill_subjects
```

Example:

```
English  
Marathi  
Hindi  
Sanskrit  
Mathematics  
EVS  
General Knowledge  
Computer  
Physical Education  
Acting
```

21. REPORT CARD PRINT FEATURES

Completed Features:

✓ Single Page Print

- ✓ No Blank Second Page
 - ✓ Search Section Hidden During Printing
 - ✓ Navigation Hidden During Printing
 - ✓ School Header Displayed Once
 - ✓ Horizontal Separator Line
 - ✓ Outer Report Card Border
 - ✓ Subject Marks Center Aligned
 - ✓ Signature Section Center Aligned
 - ✓ Passing Marks Before Obtained Marks
 - ✓ Total Obtained Marks Display
 - ✓ Larger Font Size
 - ✓ Black Header Text
 - ✓ Browser Print Button
 - ✓ Save as PDF Using Browser Print
-

22. SESSION VARIABLES USED

```
session('yearid')  
session('sectionid')  
session('year_name')  
session('section_name')
```

Helper Functions:

```
getAcademicYearIdFromSession()  
  
getOldStandardId()
```

23. IMPORTANT PROJECT FILES

Layout Files

```
resources/views/layouts/app.blade.php  
resources/views/layouts/navigation.blade.php
```

Report Card

```
resources/views/report-card/index.blade.php
```

Controllers

```
app/Http/Controllers/MarksEntryController.php  
app/Http/Controllers/Administrator/ReportCardController.php
```

24. CURRENT PROJECT STATUS

Module	Status
Authentication	Completed
Academic Year Master	Completed
Standard Master	Completed
Division Master	Completed
Subject Type Master	Completed
Subject Master	Completed
Exam Master	Completed
Student Records	Completed
Teacher Class Allocation	Completed
Teacher Subject Allocation	Completed
Skill Subject Allocation	Completed
Marks Entry	Completed

Module	Status
Marks Validation	Completed
Teacher Marks Status	Completed
Result Generation	Completed
Report Card	Completed
Print / Save PDF	Completed

END OF CURRENT ERP IMPLEMENTATION

This document represents the complete workflow, dependencies, and functionality of all modules currently implemented in the School ERP system.